

Machine Guarding SOP

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1. Purpose

This SOP defines the requirements for the design, use, inspection, and maintenance of machine guards to prevent contact with hazardous moving parts such as gears, belts, pulleys, rotating shafts, cutting blades, and points of operation. Unguarded or defeated machine guards are a leading cause of amputation and crush injuries in industrial settings.

2. Scope

Applies to all fixed, adjustable, and interlocked guards on production machinery, conveyors, presses, saws, and power transmission equipment throughout the facility.

3. Types of Machine Guards

- Fixed guards: permanently attached barriers that prevent access to hazard points and require tools to remove; used where access is not needed during normal operation.
- Interlocked guards: guards linked to the machine control such that the machine cannot operate unless the guard is closed, and opening the guard automatically stops the machine.
- Adjustable guards: provide a barrier that can be adjusted to accommodate different sizes of stock, commonly used on table saws and band saws.
- Presence-sensing devices: light curtains, safety mats, and area scanners that stop the machine when a person enters the hazard zone.
- Two-hand control devices: require simultaneous use of both hands on separate controls to initiate a machine cycle, keeping hands away from the point of operation.

4. General Requirements

Every machine with moving parts capable of causing injury must be evaluated during commissioning and assigned an appropriate guard type based on a documented risk assessment. Guards must be designed so they do not themselves create a hazard (e.g., sharp edges, pinch points) and must not obstruct required visibility or ventilation. All guards must be clearly labeled with the hazard they protect against.

5. Prohibited Practices

- Removing, bypassing, or defeating any machine guard or interlock for any reason, including increasing production speed, is strictly prohibited and is treated as a critical safety violation subject to immediate disciplinary action.
- Reaching into or around a guard to clear a jam while the machine is powered is prohibited; the machine must be de-energized and locked out first.
- Propping open interlocked guards or wedging presence-sensing devices to keep them inactive is prohibited.
- Operating any machine with a missing, damaged, or improperly reinstalled guard is prohibited.

6. Inspection Requirements

Machine guards must be visually inspected by the operator before each shift as part of the pre-operation checklist. A documented monthly inspection by the maintenance team verifies interlock function, guard integrity, and correct alignment. Any guard found damaged, missing, or malfunctioning must result in immediate machine shutdown and lockout until repaired.

7. Reporting and Corrective Action

Any employee who observes a missing, damaged, or defeated guard must immediately stop use of the machine, tag it out of service, and report it to the shift supervisor and maintenance department. Corrective action must be completed and verified before the machine is returned to service. Repeated incidents of guard defeat at a specific workstation trigger a root-cause investigation by the EHS committee.

8. Training

All operators receive machine-specific guarding training during onboarding, covering the purpose and correct use of guards on their assigned equipment, recognition of guard failure modes, and reporting procedures. Refresher training occurs annually and following any incident involving guard failure or defeat. Supervisors receive additional training on conducting guarding audits as part of the plant safety management system.